

# Michael Watters

## Scala Engineer

### Michael Watters

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Github: <https://github.com/MichaelJohnWatters>

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A Scala engineer with a background in data pipelines, custom reporting and API development. Currently building regulatory reporting for an AI-powered anti-money laundering (AML) compliance platform, submitting compliance data to government authorities in the financial crime / RegTech domain.

Experienced across the full delivery lifecycle — from writing the services in Scala and Go through Kubernetes-based deployments, GitHub Actions CI/CD, and QA automation pipelines — with a background in large-scale data processing.

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### Napier AI / Scala Engineer

January 2025 - Present, Belfast (Hybrid)

Scala engineer on Napier AI's financial crime compliance platform (AML / RegTech). Building regulatory reporting that takes compliance data through to submission to government authorities.

- Built regulatory reporting that takes compliance data end-to-end through to submission to government authorities.
- Extensive Kubernetes work — deploying and operating services on k8s.
- Built and maintained GitHub Actions CI/CD pipelines.
- Developed QA / test automation pipelines to safeguard release quality.

**Stack:** Scala, Go, Kubernetes, GitHub Actions, Kafka, PostgreSQL, Elasticsearch, SQL. QA: mabl, TestRail.

### Magnite / Software Engineer II

February 2022 - December 2025, Belfast

Data Pipeline Software Engineer building large-scale batch data pipelines to onboard and normalize advertising data, for the purpose of building advertising audiences and expanding associations using third-party identity graphs.

**Stack:** PySpark / Spark, Databricks, Python (Luigi), Snowflake, SQL, Golang APIs, Grafana and AWS services (S3, SQS, IAM, ECS, KMS, Lambda etc).

### SpotX (Acquired by Magnite) / Software Engineer I

January 2021 - February 2022, Belfast

Software Engineer, working on various custom reporting Scala applications for billing purposes and fee contracts, including a query wrapper Scala API for Apache Druid.

**Stack:** Functional Scala, Druid, Apache Kafka, Php, Jenkins, SQL, Openshift + Helm, Prometheus + Grafana. Developed with extensive CI/CD pipelines via Jenkins.

### SpotX / Trainee Software Developer

January 2019 - January 2021, Belfast

Same as Software Engineer I.

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### Programmatic Advertising Platform / Solo Project

Personal project — <https://michaeljohnwatters.github.io>

Designed and built a full programmatic advertising platform solo, mostly with the help of AI: ~170K lines of Go across 18 microservices on Kubernetes, spanning the SSP / exchange / DSP / ad-serving path, a first-party data-onboarding + identity-resolution pipeline, and an exactly-once billing ledger.

- Real-time bidding / OpenRTB auction path held at p95 ~73 ms by keeping network I/O out of the hot loop with warm in-process caches.
- End-to-end request tracing (bid → win → serve → billing) with full observability via Prometheus, Grafana, Loki and Jaeger.

**Stack:** Go, Kubernetes, Helm, PostgreSQL, ClickHouse, Redis, NATS JetStream, Kafka, S3/Parquet, Prometheus, Grafana. Domain: programmatic advertising, RTB / OpenRTB, SSP, DSP, ad serving, real-time auctions, identity resolution.

### Other

- Dart, Flutter, Flutterflow and Google Cloud Platform (Firebase etc).
- Some experience, C++ with various microcontrollers / Raspberry Pi and basic electronics, such as racing sim peripherals.
- Some experience with Java, C#, R, React, Typescript, Javascript + HTML/CSS.

## Summary

## Experience

## Projects

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## Education

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### **Ulster University** / BSc Hons Computing Systems

September 2019 - August 2022, Belfast

Awarded 1st Degree with Honors, Bachelor of Science.

### **Belfast Metropolitan College** / Foundation Degree, Software Engineering

September 2017 - June 2019, Belfast

Awarded 1st with distinction.

### **Royal Belfast Academical Institution** / A-level and GCSE

September 2005 - June 2012, Belfast

A-level, AAB (Biology, Geography and History). 8 GCSEs with Math B and English C.